PO33Q - The Life and Death of Democracies and Autocracies: A Quantitative Perspective

Marking Guidance for the Summative Research Project

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1 | Introduction

The PAIS UG Marking Criteria are built around five pillars, namely Comprehension, Analysis, Critique, Presentation and Academic Practice, and Coding. They measure the degree or quality to which each of these is executed in an assessment, using an ostensibly ordinal scale of adjectives, ranging from superlative, over excellent, very good, good, fair, limited, weak, poor, all the way to none.

Every year teaching this module, I am asked the same question: "what do I have to do to get a First?" This is a legitimate question, but there is a cognitive gap between the marking criteria and the research project itself. How does "superlative comprehension and command of the subject matter" translate into the world of a deductive research project? What does such a piece look like if you have managed to deliver an "original extension of existing arguments, debates, and interpretations"?

The marking criteria are written in general terms, while a research project is a specific piece of work with its own logic and structure. I have therefore written this document to link the marking criteria to the logic and structure of a deductive research project. In the feedback you will receive for the research project, you will find the descriptors provided in Table 4 for a more concrete explanation of how your work was marked.

> **Important**
> Before reading this document, I strongly recommend you familiarise yourself with my Guidance for Writing a Deductive Research Project.

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2 | How the Mapping Works

- Just as in the marking criteria, I have created five pillars. But my pillars represent the logic and objectives of a deductive research project: Research Conception, Scientific Coherence, Critical Interpretation, Presentation, and Coding.
- I have then divided each pillar into parts that must be present for the objective to be achieved. The parts are listed in Table 1.

Table 1: The Five Pillars of a Deductive Research Project

Pillar	The object	Its parts
Research Conception	the intellectual case for the project	puzzle → gap → research question → contribution
Scientific Coherence	the chain from theory to analysis	theory → hypotheses → attributes → measures → data → method → analysis
Critical Interpretation	the answer	report → interpret → answer → bound → place → extend
Presentation	what a competent reader can do with your project unaided	read → navigate → follow → locate → verify
Coding	the code as an instrument	execution → reproduction → implementation → economy

- The order is not arbitrary. Later parts depend on earlier ones. Each object can therefore be divided into early, middle, and late parts. A weakness early in an object is inherited by much of what follows; a weakness late in the object is more local. I have provided more detail on the parts in Section 4.
- Rather than using adjectives such as exemplary, poor, or weak, each part can only be in one of four stages of development: shown, asserted, substituted, or absent. These stages are defined in Table 2. Due to different objectives, the terminology changes across the pillars, but the underlying logic is identical.

Table 2: The Four Conditions across the Five Pillars

Condition	Research Conception	Scientific Coherence	Critical Interpretation	Presentation	Coding
Shown	derived	the connection is explicit	derived	the reader does it unaided	the code shows what it does
Asserted	asserted	the connection is implicit	asserted	the reader must supply work the project should have done	the code must be traced
Substituted	substituted	drift	substituted	the appearance of the function without the function itself	the appearance of correctness without correctness
Absent	absent	break	absent	the reader cannot do it	the code does not do it

Important
Each object must be fully constructed on the page. If one of its essential parts is omitted, the examiner has to infer it retrospectively from what the project eventually does. The merits of a project can only be assessed on what is present on the page, not by what the reader may correctly or incorrectly infer through conjecture.

3 So, what do I need to do for a First?

- To bring this back to the initial question, I can now tell you exactly what you have to do for a First.
  - You do not need better instincts, more complicated code, or more elegant prose. You need to *show* every part of the object your pillar assesses. The difference between an Upper Second and a First is the difference between doing something and demonstrating that you have done it.
  - You will note in Table 4 that within the First class, every part is already shown, so the four conditions no longer distinguish the marks. Higher Firsts are separated instead by how well the project handles the demanding cases unique to each pillar.
- Whilst striving for a First is laudable, let me also summarise what separates the other grade classes from one another:

Table 3: The Meaning of the Grade Classes

Class	Meaning
First (74–100)	Every part is shown; higher marks distinguish how well demanding cases are handled.
Upper Second (62–68)	Every part is present, but as the mark falls the first asserted part moves from late to middle to early in the object.
Lower Second (52–58)	Every part is present, but as the mark falls the first substituted part moves from late to middle to early in the object.
Third (42–48)	As the mark falls, the first absent part moves from late to middle to early in the object.
Fail (0–38)	The parts no longer constitute the object.

4 | An Explanation of the Pillars

4.1 The Research Conception Pillar

- What the marking criteria call the command of the subject matter is, in the context of this module, the degree to which you can deliver a convincing intellectual case for the research project. The intellectual case has four parts:
  1. **Puzzle:** an observation shown to violate an expectation.
  2. **Gap:** what the literature has not yet settled.
  3. **Research question:** the question that follows from the gap.
  4. **Contribution:** what answering the question would add.

- You cannot identify, state, explain, and link all of these four parts together seamlessly without a superlative or exemplary command of the subject matter. As that command becomes less pronounced through the bands, so does what you are able to build from it. A very good or considerable appreciation of the literature is usually enough to derive a puzzle and state a well-founded research question, but can leave the gap asserted rather than shown to follow from the sources reviewed. Understanding of only the core aspects of the subject matter tends to produce a rationale or a summary in place of a puzzle or a gap — the sources have been read, but not yet weighed and connected closely enough to generate either. Some or partial familiarity allows a question to be asked and even answered, but without the puzzle or gap that would explain why the answer matters. And as familiarity with the subject matter becomes little, minimal, or absent, there is progressively less from which a puzzle, a gap, a research question, or a contribution can be built.

- As shown in Table 1, these four parts form a chain. A puzzle defines the field in which the gap exists, the gap gives rise to the research question, and the contribution is what answering that question would add. Therefore, the puzzle is the early part, the gap and research question the middle part, and the contribution the late part. Early weaknesses therefore affect everything that follows. The descriptors in Table 4 record where the first weakness in this chain occurs.

- In this pillar, “showing” means “deriving”. A puzzle is derived when the expectation it violates is on the page. A gap is derived when it follows from the literature rather than being announced. A research question is derived when it follows necessarily from the identified gap. A contribution is derived when it is shown to be exactly what resolving the gap requires.

- A contribution may lie in data, methodology, theory, or any combination of these. What matters is whether the project identifies where its contribution lies and shows why advancing those dimensions is exactly what resolving the identified gap requires.

4.2 The Scientific Coherence Pillar

- You are only able to deliver "a well focused and original analysis, that has an exemplary structure with exceptional development of concepts and theories, and use of evidence" if you can show that the design of your project is perfectly coherent from theory to analysis. This pillar therefore looks at the chain that runs from the theory to the analysis:
  - theory
  - hypotheses
  - Operationalisation: attributes → measures
  - data
  - method
  - analysis
- What is described as "development of concepts and theories" in the marking criteria is what makes every decision in that chain visible as a deliberate choice, and defensible as one. As this development becomes less pronounced within the First class, the forks (see below for a definition) are increasingly less completely recognised and reasoned. At the highest levels, every fork is identified and justified; below this, the major forks remain justified while increasingly minor choices are left implicit or asserted. By the bottom of the First, the project still follows a coherent chain, but some choices are made without recognising that alternatives were available. As analytical development becomes good rather than very good or excellent in the 2.1 range, the weakness moves from the forks to the joints: the choices themselves may be defensible, but the project increasingly leaves the reader to supply why one link follows from the previous one. In the 2.2 range, those implicit joints become drift: the project moves from one link to the next, but what follows is no longer adequately determined by what came before. In the Third, drift becomes a break: one link no longer provides a defensible basis for the next. Below the pass line, there is no longer a chain that constitutes a research design, only some of its components.
- Once again, there is an order in this chain, theory → hypotheses and hypotheses → attributes are the early joints; attributes → measures and measures → data are the middle joints; and data → method and method → analysis are the late joints. A weakness at an early joint is inherited by most of the design, while a weakness at a late joint leaves more of the preceding chain intact. Below the First, the descriptors in Table 4 record where the first weakness in the chain occurs; within the First, they distinguish how completely the project's forks are recognised and justified.

> Every step of the chain must be carried through to the analysis, and every element of the analysis must trace back to the theory. Only then is the design capable of answering the research question in a fully focused way.

- When you read the descriptors in Table 4, you will notice four ideas that are unique to this pillar:
  1. **Joint:** the connection between two successive steps in the chain. A joint is explicit when the project shows why the later step follows from the earlier one; it is implicit when the connection is defensible, but left for the reader to supply.
  2. **Fork:** more than one defensible path exists, and the choice between them matters for answering the research question. The project must recognise that a choice has been made and justify the option taken. At a *major* fork, a different defensible choice would materially alter the downstream logic of the project. At a *minor* fork, it would not.
  3. **Drift:** a concept changes meaning while remaining connected. The design remains present, but tests a different claim.
  4. **Break:** one step of the chain is not carried forward. The design no longer answers the question through the theory it invokes. The pass line falls between drift and a break, as the marking criteria distinguish a very limited but still present understanding of the question, concepts, theories, and evidence at 42 from a poor grasp of the question, little knowledge of concepts and theories, and inadequate use of evidence at 38.

4.3 The Critical Interpretation Pillar

- The extent to which you can demonstrate understanding and evaluation of relevant arguments, debates, and / or interpretations and deliver clear and original extension of existing arguments, debates, and interpretations depends on six parts in the context of a research project:
  - report
  - interpret
  - answer
  - bound
  - place
  - extend

- "Regular" essays in PAIS are argued literature reviews, in which you draw on "arguments, debates, and / or interpretations" in secondary literature as your evidence. In the context of a research project with statistical analysis, the nature of the evidence changes. Here, the results tables outlining the coefficients, their significance, model fit, etc. are your primary evidence. The parts report, interpret, and answer refer to the analysis section of your research project. The parts bound, place, and extend relate to the discussion section.

- An "[e]xemplary understanding and evaluation of [evidence]" can only be demonstrated if your results are reported with exemplary clarity and complete honesty, null and insignificant results among them. Interpretation must claim exactly what the measures support, neither more nor less. Similarly, "a clear and original extension of existing arguments, debates, and interpretations" is only possible if the answer is provided, placed against the theory, and shown how it specifies the theory further. All claims must be bounded by stating the limitations of the design and what evidence would be required to close them. This will allow you to claim exactly the actual contribution you are making. With decreasing levels of capacity of independent thought, first interpretation and then bounds and placement become asserted rather than shown to follow from the evidence. Reproducing relevant arguments and debates, rather than evaluating them, corresponds to the progressive substitution of interpretation, bounds, and placement. As awareness of relevant arguments and debates becomes little or very limited, the answer and then the capacity to interpret the findings disappear, leaving findings without a discussion that makes sense of them. At the lowest levels, the findings themselves become fragmentary and eventually disappear.

- You guessed it, each part depends on the previous one. Findings must first be reported before they can be interpreted; interpretation must precede an answer; an answer must be bounded before it can be placed against the literature or extended into future research. Report and interpret are the early parts; answer and bound are the middle parts; and place and extend are the late parts. A weakness in reporting or interpretation affects the answer itself, while a weakness in placement or extension leaves the answer intact but limits what the project does with it. The descriptors in Table 4 record where the first weakness in this chain occurs.

- Three principles govern this pillar:
  1. Null results are findings and must be interpreted.
  2. Claims should match exactly what the evidence supports.
  3. Limitations bound the claim by stating what the design cannot rule out.

4.4 The Presentation Pillar

- "Presentation and Academic Practice" essentially assesses how things look from the reader's side of the page. In a research project this represents what a competent reader can do with the project unaided across five parts:
  1. read
  2. navigate
  3. follow
  4. locate
  5. verify

- An "Impeccable command of the techniques of academic writing" in the marking criteria would allow a reader to read, navigate, follow, locate, and verify the project entirely unaided, in the demanding cases as well as the routine ones. Very good use of these techniques carries a reader through the routine cases without difficulty, but the demanding ones — the figure with two scales, the source that resists easy citation — are handled less consistently. Employing only some techniques of academic writing produces the appearance of checkability without its substance: a reference list that does not actually permit location, a description of data-handling that does not permit inspection. Limited and decreasing awareness of these techniques means that the conventions of academic writing are increasingly absent, until they no longer reliably communicate what was done or why. Very poor or very weak use of them eventually leaves the project without the conventions of academic presentation in any recognisable form.

- Academic conventions exist to secure these five parts. Language allows the reader to read, structure to navigate, figures and tables to follow, references to locate, and documentation to verify.

- In this sequence, read and navigate are the early functions, follow is the middle function, and locate and verify are the late functions. This order describes dependency rather than importance: a reader cannot navigate writing they cannot read, follow an argument they cannot navigate, or verify evidence they cannot locate.

- In the previous pillars, the descriptors in Table 4 record where the first weakness in this chain occurs. But for the Presentation pillar, these parts are standing properties of the whole project rather than single acts. The descriptors therefore judge where these functions first cease to hold *consistently*.

- Within the First class, the marks are separated by how well the project handles demanding cases where the conventions alone do not determine the best solution. This translates the progression in the marking criteria from exemplary or complete command of the techniques of academic writing, through very strong and very good deployment, to good command: routine cases establish command of the conventions, while demanding cases reveal how securely that command can be applied when their application requires judgement.

4.5 The Coding Pillar

- Coding actually shares the heading with the general marking criteria, but within a research project we can identify the following four parts:
  1. execution
  2. reproduction
  3. implementation
  4. economy

- Exemplary application of coding language is what lets code not only run and reproduce the reported results, but do so in a way that is shown, on the page, to be correct and efficient, rather than simply asserted to be. Applying coding language with independent thought is usually enough to get the code running and reproducing results, but increasingly leaves its economy, implementation, or reproducibility to be taken on trust rather than demonstrated. Understanding and reproduction of coding language can produce the appearance of economy, implementation, and reproducibility without their substance: code may run and produce outputs, while the reader cannot establish that it implements the stated method or reproduces the reported analysis as claimed. Weak and decreasing competence means first reproduction, then implementation, and finally execution are lost, so progressively less of the reported analysis can actually be obtained from the code supplied. And with almost no or no understanding of coding language, there is no code that bears on the analysis at all.

- As ever, the order reflects dependency. Code that does not run cannot reproduce the results. Code whose output does not reproduce the reported results cannot function as the instrument that produced them. Code that performs the wrong method gains nothing by doing it efficiently.

- In this sequence, execution and reproduction are the early functions, implementation is the middle function, and economy is the late function. A weakness in execution or reproduction prevents the code from functioning as the instrument that produced the reported results; a weakness in economy leaves its correctness intact.

- In this pillar, *showing* means making the correspondence between code and method visible through naming, organisation and, where necessary, comments.

- Like Presentation, Coding assesses a standing property of the whole project rather than a single act. The descriptors therefore judge where these functions first cease to hold *consistently*. Only within the First class, the marks are separated by how well the code handles demanding cases where a straightforward implementation does not determine the best solution. Routine cases establish that the code executes, reproduces the results, and implements the stated method; demanding cases reveal how securely and economically those functions are maintained when the implementation itself requires judgement.

5 | Have I missed Mapping Academic Integrity?

No! The five pillars assess the quality of the work that is before the examiner. Academic integrity asks a different question: whether the work is genuinely yours to assess. Where I have reasonable grounds to think that it is not, I will refer the work to the departmental academic conduct panel.

Note that several pillars already assess practices that support academic integrity because they make it possible to scrutinise your research. In particular:

- **Presentation** asks whether a competent reader can *locate* and *verify* your work. This includes accurate attribution, sufficient referencing, a transparent account of what was done to the data, and making the code available where required.
- **Coding** asks whether the supplied code actually reproduces the reported analysis.

These are not proxies for honesty. They are scholarly conventions that make it possible to scrutinise research. They are marked on what is present in the work itself — for example, whether a source can be located or whether the reported results can be reproduced — not on suspicion about how the work was produced.

A project may therefore receive low marks for poor academic practice while raising no question of academic integrity, just as a project that raises concerns about academic integrity is dealt with through the departmental academic conduct panel rather than through these marking criteria.

These descriptors are intended to guide, not replace, academic judgement to map the logic and structure of a research project to the PAIS UG Marking Criteria. The descriptors are not intended to imply a mechanical checklist or equal weighting across dimensions; the relative significance of strengths and weaknesses may vary according to the nature, aims and design of the project.

Table 4: Mapping the Marking Criteria onto a Deductive Research Project

Class	Mark	Grade	**Research Conception & Scholarly Command** *Puzzle · research question · gap · contribution*	**Scientific Coherence & Rigour** *Theory · hypotheses · operationalisation · data · method · analysis*	**Critical Interpretation & Discussion** *Report · interpret · answer · bound · place · extend*	**Presentation & Academic Practice** *Read · navigate · follow · locate · verify*	**Coding** *Execution · reproduction · implementation · economy*
FIRST	**100**	*Excellent 1st*	The introduction opens with a genuine puzzle: the expectation that theory or intuition generates is stated, and an observation is shown to violate it. The research question follows from the puzzle and is calibrated exactly to the word limit and to the design that will answer it. The literature review is a funnel ending at that question: the field is synthesised rather than reported, the gap is identified, and what remains unresolved is specified precisely. The contribution follows from that diagnosis and is located explicitly within the research triangle. The project shows why advancing those dimension(s), whether data, methodology, theory, or their combination, is exactly what resolving the identified gap requires. Neither the puzzle nor the gap could have been stated without an exhaustive and secure command of theory and of the literature; that command is everywhere evident and nowhere asserted. In the examiner's judgement the project merits a special award.	Every step of the causal chain stated in the theory is carried through to a measure, and every element of the analysis can be traced back to a step of the chain. The concept tested is the concept the theory advanced. Where the design forks — where the theory underdetermines the choice of attribute, measure, data or method — the project recognises the fork and justifies the option taken by reference to the research goal.; where the theory closes a choice, the project shows that it does. The analysis tests the claim the theory makes, and the design is fully capable of answering the question posed. In the examiner's judgement the project merits a special award.	Findings are reported with exemplary clarity and complete honesty, null and insignificant results among them. Interpretation claims exactly what the measures support, neither more nor less. The answer is placed against the theory and shown to specify it further: from the pattern of the results, the discussion identifies the conditions under which the proposed mechanism holds and those under which it does not. Each material limitation is named, located in the design, bounded — each is shown to narrow what may be concluded — and resolved by specifying what evidence would close it; future research follows from those specifications. The question posed in the introduction is answered, and the contribution promised there is delivered. In the examiner's judgement the project merits a special award.	A competent reader can read, navigate, follow, locate and verify the project unaided, in the demanding cases as well as the routine ones, and without a single lapse. Every claim not the author's own is attributed and can be found; what was done to the data is stated, the code that did it is supplied, and everything a stranger would need to reproduce the analysis is provided. In the examiner's judgement the project merits a special award.	The code runs to completion on the data supplied and returns exactly the results the project reports. Its operations are the method the project describes, and its organisation shows that they are: the steps are named and sectioned, and commented wherever the intent is not evident from the code itself. It does what it does by the means the language provides, and the approach is suited to the size and shape of the task. Run on another machine it returns the reported tables and figures exactly; seeds are set, dependencies are declared, and a change to one step requires no change elsewhere. There is not a single lapse, in the demanding tasks as well as the routine ones. In the examiner's judgement the project merits a special award.
	94	*Excellent 1st*	The introduction opens with a genuine puzzle, its expectation stated and its anomaly shown. The question follows and is calibrated exactly to the word limit and the design. The review synthesises the field, identifies the gap, and specifies what remains unresolved. The contribution follows from that diagnosis, is located explicitly within the research triangle, and is shown convincingly to resolve the identified gap.	Every step of the causal chain is carried through to a measure, and every element of the analysis traces back to a step of the chain. The concept tested is the concept the theory advanced. Every material fork is recognised as a fork, and the option taken is justified by reference to the research goal. The analysis tests the claim the theory makes.	Findings are reported with exemplary clarity and complete honesty, null and insignificant results among them and interpreted with the same weight as positive ones. Interpretation claims exactly what the measures support. The answer is placed against the theory and used to specify and refine the conditions under which its proposed mechanism holds and does not hold. Each material limitation is admitted, traced, assessed and discharged, and future research follows from the remaining uncertainty. The question posed is answered and the promised contribution delivered.	A competent reader can read, navigate, follow, locate and verify the project unaided, in the demanding cases as well as the routine ones, and without a single lapse. Everything a stranger would need to reproduce the analysis is provided.	The code runs, reproduces the reported results exactly, implements the method described, and shows that it does. It is economical, by the means the language provides. Run on another machine it returns the reported tables and figures exactly; seeds are set, dependencies declared, and a change to one step requires no change elsewhere. There is not a single lapse, in the demanding tasks as well as the routine ones.
	88	*High 1st*	The introduction opens with a genuine puzzle. The question follows and is exactly calibrated. The review synthesises the field, identifies the gap, and specifies what remains unresolved. The contribution is located explicitly within the research triangle and is shown to resolve what the review has identified as remaining unresolved.	Every step of the causal chain is carried through to a measure, and every element of the analysis traces back to a step of the chain. The major forks are recognised and justified by reference to the research goal. At the remaining forks, a choice is made and briefly reasoned.	Findings are reported clearly and honestly, null and insignificant results among them. Interpretation claims what the measures support. The answer is placed against the theory, and the implications of the findings for its propositions are drawn out. Each material limitation is admitted, traced and assessed, and discharged by naming the evidence that would settle it; future research is motivated by that evidence rather than by the topic.	A competent reader can do all five unaided, in the demanding cases as well as the routine ones, with only isolated lapses that do not impede understanding or verification. Everything needed to reproduce the analysis is provided.	The code runs, reproduces, implements and shows that it does, in the demanding tasks as well as the routine ones, with only isolated lapses that do not affect correctness or reproducibility. The analysis is reproducible: seeds are set, dependencies are declared, and a second run returns the same results.

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Table 4: Mapping the Marking Criteria onto a Deductive Research Project (Continued)

Class	Mark	Grade	**Research Conception & Scholarly Command** *Puzzle · research question · gap · contribution*	**Scientific Coherence & Rigour** *Theory · hypotheses · operationalisation · data · method · analysis*	**Critical Interpretation & Discussion** *Report · interpret · answer · bound · place · extend*	**Presentation & Academic Practice** *Read · navigate · follow · locate · verify*	**Coding** *Execution · reproduction · implementation · economy*
FIRST	**82**	*Upper Mid 1st*	The introduction opens with a genuine puzzle. The question follows and is well calibrated. The review synthesises the field and identifies the gap. The contribution is located within the research triangle and is shown to address that gap.	The chain is complete in both directions. At each fork a choice is made and a reason for it is given. Where obvious alternatives exist, the project does not show why the chosen option serves the research goal better.	Findings are reported clearly and interpreted accurately, null results included. The answer is placed against the theory, and what the findings strengthen and what they weaken in it is drawn out. Each material limitation is admitted, traced to the design and assessed: each is shown to narrow what may be concluded. Future research follows from the remaining uncertainty.	A competent reader can do all five unaided across the demanding cases as well as the routine ones. The demanding cases are handled deliberately, with only occasional lapses among them.	The code runs, reproduces, implements and shows that it does, in the demanding tasks as well as the routine ones, with occasional isolated lapses that do not affect correctness or reproducibility. The approach chosen suits the size and shape of the task.
	78	*Lower Mid 1st*	The introduction opens with a genuine puzzle and the question follows from it. The review synthesises the field and identifies the gap. The contribution is located within the research triangle, but it is not explicitly shown how advancing those dimension(s) addresses the identified gap.	The chain is complete in both directions. The major forks are reasoned; at the remaining forks the choice is made explicitly, but asserted rather than argued.	Findings are reported clearly and interpreted accurately. The answer is placed against the theory: it is shown to support, to qualify, or to fail to support the proposed mechanism. Each material limitation is admitted and traced to the design, though its cost to the claim is not assessed. Future research follows from the remaining uncertainty.	A competent reader can do all five unaided in the routine cases. The demanding cases are handled too, but the reader feels the effort in them: the two-scale figure is legible only on a second look, the awkward source cited in a form that just about works.	The code runs, reproduces, implements and shows that it does. The demanding tasks, such as a merge on duplicate keys, missing data, or a computation that must be vectorised, are handled deliberately, with isolated lapses among them: a hard-coded path, say, or an object created and never used.
	74	*Low 1st*	The introduction opens with a genuine puzzle and the question follows from it. The review identifies the gap. A contribution is made within the research triangle, but the project identifies only part of its contribution or misidentifies where its principal contribution lies. The reader can see a broader or different contribution than the project itself recognises.	The chain is complete in both directions. The choices of attribute, measure, data and method are stated and applied consistently, but the project treats them as though they admitted no alternatives: the reader cannot tell that the forks were recognised as forks.	Findings are reported clearly and interpreted accurately. The answer is given, and each material limitation is admitted and traced to the design, though its cost to the claim is not assessed. The answer is returned to the literature: the finding is placed against the studies reviewed. What it does to the theory is stated but not pursued, and the discussion is accurate rather than analytical.	A competent reader can do all five unaided in the routine cases. The demanding cases are met unevenly, and the project does not appear to distinguish them from the routine ones: some hard cases are handled, others slip, with no sign the writer saw them as hard.	The code runs, reproduces, implements and shows that it does, throughout the routine tasks. The demanding tasks succeed unevenly: they work, but by brute force, or because the data happen to be small, and the project does not appear to distinguish them from the routine ones.
UPPER SECOND (2.1)	**68**	*High 2.1*	Puzzle, question and gap are all derived. The contribution is asserted rather than derived: it is claimed in the introduction, and may be located within the research triangle, but the reader is not shown why advancing those dimension(s) is what the identified gap requires.	Every step of the chain is carried through to a measure and every element of the analysis traces back to it, but the first implicit joint occurs late in the chain. The reader can see, for example, that the analysis follows from the method, though the project does not say so.	Report, interpretation, answer and bounds are all derived. Placement is asserted rather than derived: the discussion states that the finding matters to the field, but does not show what it changes. Future research follows from the findings.	Reading, navigation and following remain possible without assistance. The first difficulty occurs in locating and verifying specific claims, which now require work the project should have done.	The code runs, reproduces the reported results, implements the method, and its steps are named. The first weakness occurs in economy: where the approach mattered — for example a loop or a vectorised idiom, one merge strategy or another — nothing shows why this one was chosen, and it must be taken on trust.
	65	*Mid 2.1*	The puzzle is derived and the research question follows from the identified gap. The gap is asserted rather than derived: the review weighs and connects its sources and arrives at a gap, but does not show that the gap follows from them. The contribution is located within the research triangle, but rests on the gap as asserted.	Every step of the chain is carried through to a measure and every element of the analysis traces back to it, but the first implicit joint occurs in the middle of the chain. For example, it can be reconstructed what the measures are taken to capture, but this is not stated anywhere.	Findings are reported and interpreted accurately, and the question is answered. Bounds are asserted rather than derived: material limitations are admitted but their cost is not assessed, and the reader must work out for themselves what the answer may no longer support.	Reading and navigation remain possible without assistance. The first difficulty occurs in following the argument. Figures and tables can be understood only by hunting through the prose for their explanation, because the captions do not stand alone and the text does not say where to look.	The code runs and returns the reported results. The first weakness occurs in legibility: that its operations implement the method described can be established only by tracing the script line by line. The steps are not named, not sectioned, and not commented where the intent is not evident. It is correct, and the reader must prove it.
	62	*Low 2.1*	The research question is clearly stated and well calibrated. The puzzle is asserted rather than derived: an observation is presented as surprising without the expectation it violates being stated, so the reader must supply the theory that makes it a puzzle. The gap and the contribution, however well located within the research triangle, rest on a puzzle the project has not made.	Every step of the chain is carried through to a measure and the analysis traces back to it, but the first implicit joint occurs early in the chain. Everything downstream rests on a step — such as the derivation of the hypotheses from the theory — that the reader must supply unaided.	Findings are reported. Interpretation is asserted rather than derived: the results are said to support or to fail to support the hypotheses, but the reader must go to the tables to see that they do. The answer, its bounds and its placement all rest on a reading of the evidence that the project has not performed on the page.	Reading remains possible without assistance. The first difficulty occurs in navigation. The structure does not signal where things are, and lapses of spelling and grammar oblige repeated rereading. Everything downstream is read through that effort.	The first weakness occurs in execution. The code runs only after the reader supplies what the project omitted: an installation, a file path, an order of execution, or a seed. Everything downstream depends upon that repair.

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Table 4: Mapping the Marking Criteria onto a Deductive Research Project (Continued)

Class	Mark	Grade	**Research Conception & Scholarly Command** *Puzzle · research question · gap · contribution*	**Scientific Coherence & Rigour** *Theory · hypotheses · operationalisation · data · method · analysis*	**Critical Interpretation & Discussion** *Report · interpret · answer · bound · place · extend*	**Presentation & Academic Practice** *Read · navigate · follow · locate · verify*	**Coding** *Execution · reproduction · implementation · economy*
LOWER SECOND (2.2)	**58**	*High 2.2*	The puzzle is asserted and the gap identified. What is claimed as a contribution is the default one, presented as though it were substantive: the data are said to be recent, or the sources newly combined. The project locates its contribution within the research triangle, but mistakes novelty of material or assembly for an advance in data, methodology, or theory. The claim is true, and it is worth nothing.	The concept drifts at a late joint. The analysis treats the variables as though they captured somewhat more than they do, so the claim tested is slightly wider than the measures support. The design still answers the question, but the answer is less secure than the project presents it as being.	Report, interpretation, answer and bounds are all derived. The first weakness occurs in placement: the discussion largely restates the findings rather than drawing out their implications for the theory or the literature.	The first failure occurs in verification. The apparatus of checkability is present in form only. There is a reference list, and it does not permit location; there is an account of the data, and it says that they were cleaned without saying how; there is a statement of availability, and the data cannot be obtained. The project has the shape of checkable work and cannot be checked.	The code has the appearance of economical work without its effect. Economy is asserted rather than realised: a vectorised idiom is used in one place and a loop for the same job in another, or a function is defined and called only once. It runs, reproduces and implements the method; only its economy is for show.
	55	*Mid 2.2*	The puzzle is asserted and the research question follows from the identified gap. In place of a gap the review offers a summary: sources are reported one by one rather than weighed and connected, and a gap is announced at the end rather than shown to follow from them. What is presented as command of the literature is acquaintance with it. The contribution rests on the announced gap.	The concept drifts at a middle joint. The measures do not climb back to the attributes they were chosen for, so the concept that reaches the analysis is not the concept the theory advanced. Every subsequent part of the project inherits the alteration.	Report, interpretation and answer are derived. The first weakness occurs in bounding: in place of limitations the project offers controllable constraints — the time was short, the word limit was binding — rather than features of the design that limit what may be concluded.	The first failure occurs in locating the evidence on which particular claims depend. Figures and tables are numbered and captioned, but units and sources are missing, and no figure is called upon where the argument needs it. They decorate the analysis rather than carry it.	Implementation is asserted rather than realised. A step that purports to derive its values hard-codes them, for example, or a filter drops observations the method retains. The code runs, but it is not doing what the project says was done.
	52	*Low 2.2*	In place of a puzzle the introduction offers a rationale: the subject is said to be important, or timely, or to have gone unexamined. The question is stated and answerable, and the reader is told why it may be asked, but not why its answer should surprise anyone. Where the rationale is that nobody has asked before, the gap has been made to do the puzzle's work, and it cannot: it establishes that the question is unanswered, not that it is worth answering.	The concept drifts at an early joint. The hypotheses do not isolate the propositions of the causal chain, and test a weaker or a wider claim than the theory advances. The whole of the design downstream inherits the defect.	Report is derived. The first weakness occurs in interpretation: the results are narrated a second time in prose, but what they mean for the hypotheses is never shown. An answer is reached, but not from the evidence set out.	The first failure occurs in following the argument. The sections bear the expected names but not the expected content: headings do not correspond to the argument beneath them, and the reader is directed to places the project does not go.	Reproducibility is asserted rather than realised. The code produces output that has the appearance of the reported results without being confirmably them: a seed is set after the random numbers are drawn, or an intermediate object is assumed, so a second run need not return the same numbers. It looks reproducible and is not.

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Table 4: Mapping the Marking Criteria onto a Deductive Research Project (Continued)

Class	Mark	Grade	**Research Conception & Scholarly Command** *Puzzle · research question · gap · contribution*	**Scientific Coherence & Rigour** *Theory · hypotheses · operationalisation · data · method · analysis*	**Critical Interpretation & Discussion** *Report · interpret · answer · bound · place · extend*	**Presentation & Academic Practice** *Read · navigate · follow · locate · verify*	**Coding** *Execution · reproduction · implementation · economy*
THIRD	**48**	*High 3rd*	No contribution is claimed. The project makes the default contribution that any competent reading would have made, and never recognises it, or says what an answer would add within the research triangle. A question is asked, and a gap is identified; what the field gains by answering it is left for the reader to supply.	The first break in the chain occurs at a late joint: for example, the analysis does not follow from the method. The criteria by which the evidence is assessed are not derived from the theory, so the procedure yields description rather than a test of the hypotheses. The joints before it hold, but the question is not answered by what was done.	Report, interpretation, answer and bounds are derived. The first missing step is placement: the findings are never returned to the theory or to the literature from which the project began, and nothing follows from them. The project reaches an answer and stops.	The conventions of academic presentation no longer secure verification. References cannot be traced, the handling of the data cannot be inspected, and the analysis cannot be reproduced.	The first failure occurs in reproduction. The reported results cannot be obtained from the code. It runs and implements the method, but re-running it, or running it on another machine, does not return the numbers the project reports, and no step accounts for the difference.
	45	*Mid 3rd*	No gap is identified. The literature is cited, but the reader is never shown that the question is unanswered, or that the field has failed to settle it. Whatever the project contributes therefore floats free, since nothing has been established for it to fill.	The first break in the chain occurs at a middle joint: for example, a step of the causal chain has no measure, or a measure answers to no attribute. The theory and hypotheses are present, but the test downstream is not meaningful.	Report, interpretation and answer are derived. The first missing step is bounding: no material limitations are identified, and the answer is presented as though the design could not have gone wrong. The reader is given no way to judge what the project has failed to rule out.	The first failure occurs in presenting the evidence itself. Figures have no axes, labels or units; tables have no headings. What was found is asserted in the prose and nowhere shown.	The first failure occurs in implementation. The code runs and returns output, but it does not implement the method the project describes. What is reported was produced by something other than what the project says was done.
	42	*Low 3rd*	The puzzle is absent. No expectation is offered and no observation is shown to violate one, so nothing is at stake in the answer. A research question is stated and could in principle be answered, and a gap may be identified and a contribution claimed; but a gap only shows the answer is missing, where a puzzle would show it is worth answering — and an unanswered question is not the same as one worth answering.	The first break in the chain occurs at an early joint: for example, the hypotheses do not follow from the theory. What the project does is legible as research, but the question is not evaluated through the lens of the theory it invokes, and so is not answered.	Report is derived. The first missing step is interpretation: findings are reported, and an answer to the research question is asserted, but nothing shows that the answer follows from what was found.	The first failure occurs in communicating the research report as such. The reader can still recognise the project as an attempt at academic research, but can no longer recover with confidence what was done or why. The conventions are attempted throughout, yet the writing obstructs sense and the structure no longer communicates the intended argument.	The first failure occurs in execution. The code does not run on the data supplied. It covers the analysis, but the results the project reports cannot be obtained by executing it.
FAIL	**38**	*High Fail*	No intellectual case for the project is made. A topic is announced and the literature is cited and engaged, but no coherent case is developed for why a particular question should be asked, so the project remains a topic rather than a research project.	Every part of the project is present, and connections between the parts are claimed, but no joint holds. The project is a set of research components rather than a research design.	Report is derived. The first missing step is answering the research question: findings are reported correctly, but no answer is given. Nothing is interpreted, discussed or bounded.	The conventions of academic presentation are no longer consistently attempted. Only some are in evidence anywhere in the project, and the work is no longer recognisable as a coherent academic research report.	The code accounts for only some of the analysis reported.
	32	*Fail*	A topic is announced. Sources are named but not engaged, and no theory or analytical framework is identified.	Most parts are present but no connection between them is claimed. Theory, data and analysis stand side by side without reference to one another.	Findings are reported, but not in a form from which the reader can tell what was found. No answer is given and there is no discussion.	Few of the conventions of academic presentation are in evidence.	The code accounts for few of the steps of the analysis.
	25	*Low Fail*	The topic is barely identifiable. There is almost no engagement with the literature, and no theory.	Some parts are absent — for example there is no operationalisation, or no method — and those present are unconnected.	Findings are absent or incoherent. There is no discussion.	Very few of the conventions of academic presentation are in evidence.	The code accounts for very few of the steps of the analysis.
	12	*Very Low Fail*	No topic is discernible, and almost no engagement with any literature or theory.	Almost none of the parts of a research design is present.	There are no findings and no interpretation.	Almost none of the conventions of academic presentation is in evidence.	The code accounts for almost none of the analysis.
ZERO	**0**	*Zero*	No engagement with an intellectual case for the project, or with relevant theory and literature.	No part of a research design is present.	No results, interpretation, or discussion.	None of the conventions of academic presentation is in evidence.	No code is supplied, or the code supplied has no bearing on the project.

7 | Mapping of Marks: Quick Reference

- Table 5 provides a quick overview of the descriptors. Each cell names the condition that characterises that grade. It is intended as a reference aid for both students and markers. The full descriptors in Table 4 remain authoritative.

Table 5: Mapping of Marks – Quick Reference

Mark	Research Conception	Scientific Coherence	Critical Interpretation	Presentation	Coding
100	Special award	Special award	Special award	Special award	Special award
94	Every part exemplary	Every material fork justified	Every material limitation discharged	Demanding cases flawless	Demanding cases flawless
88	Contribution fully shown	Major forks justified	Theory refined	Only isolated lapses	Only isolated lapses
82	Contribution shown	Forks recognised and reasoned	Theory implications developed	Demanding cases handled	Approach suited to task
78	Contribution link not fully shown	Minor forks asserted	Theory tested	Demanding cases effortful	Demanding cases effortful
74	Contribution partly misidentified	Forks not recognised as forks	Theory returned to	Demanding cases uneven	Demanding cases uneven
68	Contribution asserted	Late joint implicit	Placement asserted	Locate/verify asserted	Economy asserted
65	Gap asserted	Middle joint implicit	Bounds asserted	Following asserted	Implementation traceable
62	Puzzle asserted	Early joint implicit	Interpretation asserted	Navigation asserted	Execution repairable
58	Default contribution substituted	Late drift	Placement substituted	Verification substituted	Economy substituted
55	Summary substitutes for gap	Middle drift	Constraints substitute for bounds	Location substituted	Implementation substituted
52	Rationale substitutes for puzzle	Early drift	Restatement substitutes for interpretation	Following substituted	Reproducibility substituted
48	Contribution absent	Late break	Placement absent	Verification absent	Reproduction absent
45	Gap absent	Middle break	Bounds absent	Evidence presentation absent	Implementation absent
42	Puzzle absent	Early break	Interpretation absent	Research report fails	Execution absent
38	Intellectual case disappears	Design disappears	Answer disappears	Conventions disappear	Analysis code disappears
32	Sources not engaged	Components only	Findings only	Few conventions	Few analysis steps coded
25	Barely identifiable topic	Fragmentary design	Fragmentary findings	Very few conventions	Very little analysis coded
12	No discernible topic	Almost no design	No findings	Almost no conventions	Almost no analysis coded
0	No research conception	No design	No results or discussion	No academic presentation	No relevant code